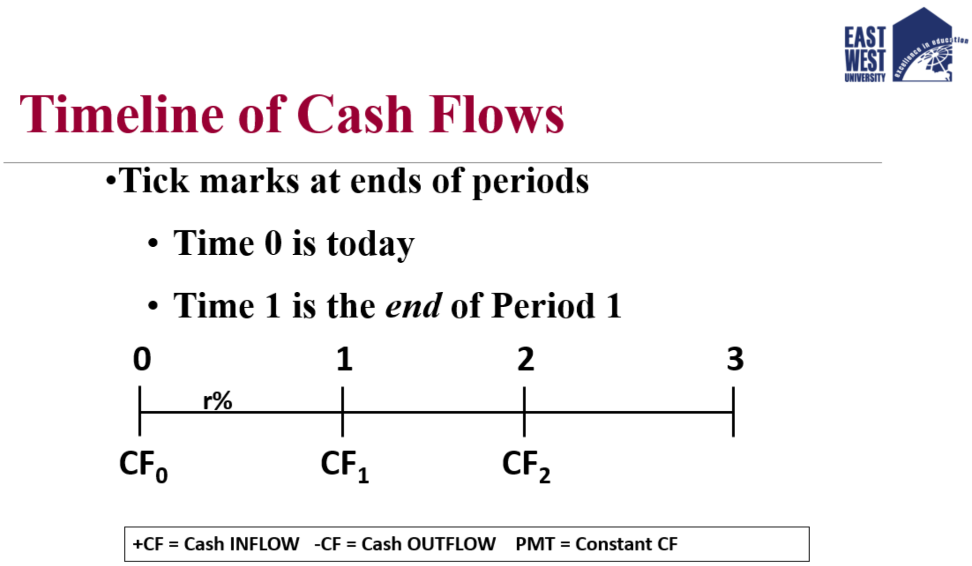
Recommended Reading:
Scott Besley and Eugene F. Brigham – Essentials of Managerial Finance, 14th Edition, The Dryden Press, 2008.

Chapter 4 – Time Value of Money

- **Time value of money** (A dollar in hand today is worth more than a dollar promised at some time in the future).

- Timeline of Cash Flows



- Present value (PV)
The current value of future cash flows discounted at the appropriate discount rate.
Value at t=0 on a timeline

- Future value (FV)
The amount an investment is worth after one or more periods. 'Later' money on a timeline

- Interest rate (r)
Discount rate
Cost of capital
Opportunity cost of capital
Required return

Future Values: Example 1

Investing for a single period

- Suppose you invest $100 for one year at 10% (r) per year. What is the future value in one year?
  - Interest = 100(.1) = $10
  - Value in one year = principal + interest = 100 + 10 = $110
  - Future value (FV) = 100(1 + .1) = $110

Investing for more than one period

- Suppose you leave the money in for another year. How much will you have two years from now?
  - FV = 100(1.1)(1.1) = 1000(1.1)^2 = $121

Future Values: General Formula

$FV = PV(1 + r)^t$

- FV = Future value
- PV = Present value
- r = Period interest rate, expressed as a decimal
- t = Number of periods

- Simple interest

- Compound interest

Future values: Effects of compounding

Consider the previous example:

- FV - simple interest
  = 100 + 10 + 10 = 120
- FV - compound interest
  $=100(1.10)^2 = 121.00$
- The extra 1.00 comes from the interest of .10(10) = 1.00 earned on the first interest payment.

Future value: Example 3

Suppose you had a relative deposit $5 at 6% interest 200 years ago. How much would the investment be worth today?

- $FV = 5(1.06)^{200} = \$575\ 629.50$

What is the effect of compounding?

- Simple interest = 5 + 200(5)(.06) = 5+60=65
- Compounding added $575564.5 to the value of the investment.

Present value

$$FV = PV(1 + r)^t$$

Rearrange to solve for basic present value equation

$$PV = FV / (1+r)^t$$
$$PV = FV(1+r)^{-t}$$

'Discounting' = finding the present value of one or more future amounts.

Present value
Example 1—Single period

Suppose you need $400 to buy textbooks next year. You can earn 7% on your money. How much do you have to put up today?

- Formula solution:
  - $PV=FV(1+r)^{-t}=400/(1+0.07)$
  - PV= 373.83

Present value
Example 2—Multiperiod

You would like to buy a new car. You have $50 000, but the car costs $68 500. If you can earn 9%, how much do you have to invest today to buy the car in two years? Do you have enough? Assume the price will stay the same.

- $PV=FV/(1+r)^{t}$
- $=68500/(1.09)^{2}$
- 57655.08, so still short of the required amount by $7655

Present vs future value—
Evaluating investments

Your company proposes to buy an asset for $335. This investment is very safe. You will sell the asset in 3 years for $400. You know you could invest the $335 elsewhere at 10% with very little risk. What do you think of the proposed investment?

Discount rate

To find the implied interest rate, rearrange the basic PV equation and solve for r:

$FV = PV(1 + r)^t$

$\Rightarrow FV/PV = (1 + r)^t$

$\Rightarrow \sqrt[t]{FV/PV} = \sqrt[t]{(1 + r)^t}$

$\Rightarrow FV/PV^{1/t} = 1 + r$

$\Rightarrow FV/PV^{1/t} = 1 + r$

$\Rightarrow FV/PV^{1/t} - 1 = r$

$\Rightarrow$ **r = (FV / PV)$^{1/t}$ – 1**

Discount rate:

Suppose you are offered an investment that will allow you to double your money in 6 years. You have $10 000 to invest. What is the implied rate of interest?

Finding the number of periods

Start with basic equation and solve for t:

$FV = PV(1 + r)^t$

$FV/PV = (1 + r)^t$

$(1 + r)^t = FV/PV$

$t \ln(1+r) = \ln(FV/PV)$

$t = \ln(FV/PV)/ \ln(1+r)$

$t \ln(1+r)/ \ln(1+r) = \ln(FV/PV)/ \ln(1+r)$

$t = \ln(FV/PV)/ \ln(1+r)$

$$\boxed{t = \frac{\ln\left(\frac{FV}{PV}\right)}{\ln(1+r)}}$$

Number of periods: Example 1

You want to purchase a new car and you are willing to pay $20 000. If you can invest at 10% per year and you currently have $15 000, how long will it be before you have enough money to pay cash for the car?

Present value: Multiple cash flows
Example

You are offered an investment that will pay:

- $200 in year 1;
- $400 the next year;
- $600 the following year; and
- $800 at the end of the 4th year.

You can earn 12% on similar investments. What is the most you should pay for this one?

Find the PV of each cash flow and add them:

- Year 1 CF: $\$200 / (1.12)^1 = \$\ 178.57$
- Year 2 CF: $\$400 / (1.12)^2 = \$\ 318.88$
- Year 3 CF: $\$600 / (1.12)^3 = \$\ 427.07$
- Year 4 CF: $\$800 / (1.12)^4 = \$\ 508.41$
- Total PV = $1432.93

Future value: Multiple cash flows Example

You deposit $100 in one year, $200 in two years and $300 in three years.

How much will you have in 3 years at 7% interest?

- Year 1: $FV = \$100(1.07)^2 = \$\ 114.49$
- Year 2: $FV = \$200(1.07)^1 = \$\ 214.00$
- Year 3: $FV = \$300 = \$\ 300.00$
- Total value in 3 years = $628.49

✓Problem: 4·13 (Page 168 – recommended book)

- Annuities and perpetuities

✓ Annuity—finite series of equal payments that occur at regular intervals

✓ If the first payment occurs at the end of the period, it is called an ordinary annuity

✓ If the first payment occurs at the beginning of the period, it is called an annuity due

✓ Perpetuity—infinite series of equal payments

Annuities and perpetuities

Basic formulas

Perpetuity: PV = C/r

Annuities:

$$PV = C\left[\frac{1 - \frac{1}{(1+r)^t}}{r}\right]$$

$$FV = C\left[\frac{(1+r)^t - 1}{r}\right]$$

Future values for annuities

Suppose you begin saving for your retirement by depositing $2000 per year in a superannuation fund. If the interest rate is 7.5%, how much will you have in 40 years?

$$\boxed{\begin{aligned} FV &= PMT\left[\frac{(1+r)^t - 1}{r}\right] \\ FV &= -2000\left[\frac{(1.075)^{40} - 1}{.075}\right] = 454{,}513.04 \end{aligned}}$$

Calculating perpetuity values

Dawa Financial is trying to sell you an investment policy that will pay you and your heirs $35,000 per year forever. If the required return on this investment is 7 percent, how much will you pay for the policy?

To find the PV of a perpetuity, we use the equation:

$PV_\infty = C / r$ *or* $PV_\infty = PMT / i$

$= \$35{,}000 / .07$
$= \$500{,}000$

Problem: 4·13 Page 168

Find the amount to which $500 will grow in five years under each of the following conditions:

a. 12 percent compounded annually.

b. 12 percent compounded semiannually.

c. 12 percent compounded quarterly.

d. 12 percent compounded monthly.

I. Symbols

PV = Present value, what future cash flows are worth today
FV_t = Future value, what cash flows are worth in the futurer
r = Interest rate, rate of return, or discount rate per period—typically, but not always, one year
t = Number of periods—typically, but not always, the number of years
C = Cash amount

II. Future value of *C* invested at *r*% per period for *t* periods

$FV_t = C \times (1 + r)^t$
The term $(1 + r)^t$ is called the *future value factor.*

III. Present value of *C* to be received in *t* periods at *r*% per period

$PV = C / (1 + r)^t$
The term $1/(1 + r)^t$ is called the *present value factor.*

IV. The basic present value equation giving the relationship between present and future value is:

$PV = FV_t/(1 + r)^t$

I. Symbols

PV = Present value, what future cash flows are worth today

FV_t = Future value, what cash flows are worth in the future at time t

r = Interest rate, rate of return, or discount rate per period—typically, but not always, one year

t = Number of periods—typically, but not always, the number of years

C = Cash amount

II. Future value of C invested per period fortperiods atr% per period

$FV_t = C \times [(1 + r)^t - 1]/r$

A series of identical cash flows is called an annuity, and the term $[(1 + r)^t - 1]/r$ is called the *annuity future value factor.*

III. Present value of C per period fortperiods atr% per period

$PV = C \times \{1 - [1/(1 + r)^t]\}/r$

The term $\{1 - [1/(1 + r)^t]\}/r$ is called the *annuity present value factor.*

IV. Present value of a perpetuity of C per period

$PV = C/r$

A perpetuity has the same cash flow every year forever.

Loan Amortization

- Example from book (Page 161, Table 4-1)
- Require equal payments over its life
- Payments include both interest and repayment of the debt.

Amortized loan with fixed payment— Example

Each payment covers the interest expense plus reduces principal.

Consider a 5-year loan with annual payments. The interest rate is 9% and the principal amount is $5000.

- What is the annual payment?
  - $5000 = PMT[1 - 1 / 1.09^5] / .09$ ➔PMT = 1285.46
  - =PMT(0.09,5,5000,0) = 1285.46

****Please study the concepts in detail and practice the mathematical problems from the recommended book. It will not be much helpful in the exam if you follow my slides only****